Alternate method:

Using the formula (6.23)

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=-\frac{(ax_1+by_1+c)}{a^2+b^2}$$
$$\frac{x-0}{2}=\frac{y-0}{1}=-\frac{(2(0)+1(0)-5)}{2^2+1^2}$$
$$\Rightarrow \frac{x}{2}=\frac{y}{1}=1\Rightarrow(2,\,1)$$

Example 6.25 Find the equation of the bisector of the acute angle between the lines $3x+4y+2=0$ and $5x+12y-5=0$.

Solution:

First, let us make the constant term positives in both the equations.

The angle bisectors of the given equations are

$$\frac{3x+4y+2}{\sqrt{3^2+4^2}}=\pm\frac{-5x-12y+5}{\sqrt{5^2+12^2}}\quad\text{(moving point is equidistance from the lines)}$$

Since $a_1a_2+b_1b_2=-15-48<0$, the equation of bisector of the acute angle between the lines is $\frac{3x+4y+2}{5}=+\frac{-5x-12y+5}{13}\Rightarrow 64x+112y+1=0$

Example 6.26 Find the points on the line $x+y=5$, that lie at a distance 2 units from the line $4x+3y-12=0$

Solution:

Any point on the line $x+y=5$ is $x=t, y=5-t$

The distance from $(t, 5-t)$ to the line $4x+3y-12=0$ is given by 2 units.

$$\therefore\ \frac{4(t)+3(5-t)-12}{\sqrt{4^2+3^2}}=2$$
$$\Rightarrow\frac{|t+3|}{5}=2$$
$$\Rightarrow t+3=\pm 10\Rightarrow t=-13,\ t=7$$

$\therefore$ The points $(-13, 18)$ and $(7, -2)$.

Example 6.27 A straight line passes through a fixed point $(6,\ 8)$. Find the locus of the foot of the perpendicular drawn to it from the origin O.

Solution:

Let the point (x_1, y_1) be (6, 8). and $P(h, k)$ be a point on the required locus.

Family of equations of the straight lines passing through the fixed point (x_1, y_1) is

$$y-y_1=m(x-x_1)\ \Rightarrow\ y-8=m(x-6)\qquad(6.25)$$